

Experiment 6 Windows Tabs and Iframes

Objective: learn how browser automation changes context when an element is inside an iframe and when a link opens another browser tab or window.

1 Concepts

Iframe context: Selenium cannot locate elements inside an iframe until the driver switches into that frame. Use `switch_to.frame(...)`, then return with `switch_to.default_content()`. Window context: save the original handle, inspect `window_handles`, switch to the new handle, then switch back before quitting.

2 Robust implementation

```
import time

from selenium import webdriver

from selenium.webdriver.common.by import By

from selenium.webdriver.support.ui import WebDriverWait

driver = webdriver.Chrome()

wait = WebDriverWait(driver, 10)

try:

    driver.get("https://rahulshettyacademy.com/AutomationPractice/")

    main = driver.current_window_handle

    frame = wait.until(lambda browser: browser.find_element(By.CSS_SELECTOR, "iframe"))

    driver.switch_to.frame(frame)

    iframe_text = driver.find_element(By.TAG_NAME, "body").text[:300]

    print("Iframe text:", repr(iframe_text))

    driver.switch_to.default_content()

    before = set(driver.window_handles)
```

```

driver.find_element(By.ID, "opentab").click()

wait.until(lambda browser: len(browser.window_handles) > len(before))

child = next(handle for handle in driver.window_handles if handle != main)

driver.switch_to.window(child)

print("Child title:", driver.title)


driver.close()

driver.switch_to.window(main)

print("Main title:", driver.title)

finally:

    time.sleep(10)

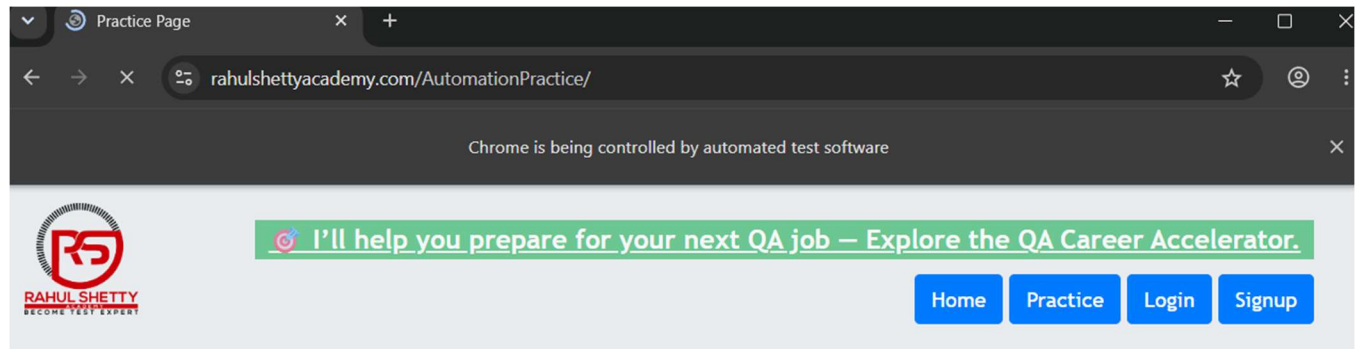
    driver.quit()

```

3 Live test

Check	Observed result	Status
Main page	Practice Page loaded at rahulshettyacademy.com/AutomationPractice/	PASS
Iframe present	Iframe and embedded practice content were visible.	PASS
Old contact-email assertion	The PDF expected contact@... text was not exposed by the live iframe today.	CHANGED
Open Tab control	Click did not create a second tab in this in-app browser session.	CHANGED
Child URL check	qaclickacademy.com returned Cloudflare Invalid SSL certificate (526).	BLOCKED

4 Screenshot evidence



Practice Page

Radio Button Example

☐ Radio1
☐ Radio2
☐ Radio3

Suggestion Class Example

Type to Select Countries

Dropdown Example

Select ▼

Checkbox Example

☐ Option1
☐ Option2
☐ Option3

Switch Window Example

Open Window

Switch Tab Example

Open Tab

Switch To Alert Example

Enter Your Name

Alert Confirm

Web Table Example

Element Displayed Example

5 Conclusion

The procedure adds waits and defensive handle selection. The main page and iframe are available, but the old iframe text and external tab destination are currently unstable. Update the locator or assertion when the site changes.